

32. A box is 15 centimeters long and 5 centimeters wide. The volume of the box is 225 cm^3 . What is the height of the box?
33. Kareem mixed two cartons of orange juice, three 2-liter bottles of soda water and six cans of cocktail fruits to make a fruit punch for a party. The cartons of orange juice and cans of cocktail fruits each have a volume of 500 cm^3 . How much punch, in liters, did Kareem make?
34. A holding tank has dimensions of $16 \text{ m} \times 8 \text{ m} \times 8 \text{ m}$. If the tank is half-full, how more liters of liquid can the tank hold?
35. A large plastic storage bin has dimensions of $16 \text{ cm} \times 16 \text{ cm} \times 16 \text{ cm}$. A medium bin's dimensions are half the size of the large bin. A small bin's dimensions are the size of the medium bin. If the storage bins come in a set of 3—small, medium, and large—what is the total volume of the storage bin set in cubic centimeters?
36. A soft serve ice cream machine holds a 19.2 liter bag of ice cream mix. If the average serving size of an ice cream cone is 120 mL, how many cones can be made from each bag of mix?
37. A shipping carton has dimensions of $0.5 \text{ m} \times 0.5 \text{ m} \times 0.5 \text{ m}$. How many boxes with dimensions of $50 \text{ mm} \times 50 \text{ mm} \times 50 \text{ mm}$ will fit in the shipping carton?
38. A recipe for chili makes 3.5 liters of chili. If a restaurant serves chili in 250 mL bowls, how many bowls of chili can they serve?
39. A contractor is building an in-ground pool. They excavate a pit that measures $12 \text{ m} \times 9 \text{ m} \times 2.5 \text{ m}$. The dirt is being taken away in a truck that holds 30 m^3 . How many trips will the truck have to make to cart away all of the dirt?
40. A juice dispenser measures $30 \text{ cm} \times 30 \text{ cm} \times 30 \text{ cm}$. How many 375 mL servings will a full dispenser serve?

9.4 Measuring Weight



Figure 9.12 Weight scale at the local Antigua market (credit: “Weight scale at the local Antigua market” by Lucía García González/Flickr, CC0 1.0 Public Domain Dedication)

Learning Objectives

After completing this section, you should be able to:

1. Identify reasonable values for weight applications.
2. Convert units of measures of weight.
3. Solve application problems involving weight.

In the metric system, weight is expressed in terms of grams or kilograms, with a kilogram being equal to 1,000 grams. A paper clip weighs about 1 gram. A liter of water weighs about 1 kilogram. In fact, in the same way that 1 liter is equal in volume to 1 cubic decimeter, the kilogram was originally defined as the mass of 1 liter of water. In some cases, particularly in scientific or medical settings where small amounts of materials are used, the milligram is used to express weight. At the other end of the scale is the metric ton (mt), which is equivalent to 1,000 kilograms. The average car weighs about 2 metric tons.

Any discussion about metric weight must also include a conversation about **mass**. Scientifically, mass is the amount of

matter in an object whereas weight is the force exerted on an object by gravity. The amount of mass of an object remains constant no matter where the object is. Identical objects located on Earth and on the moon will have the same mass, but the weight of the objects will differ because the moon has a weaker gravitational force than Earth. So, objects with the same mass will weigh less on the moon than on Earth.

Since there is no easy way to measure mass, and since gravity is just about the same no matter where on Earth you go, people in countries that use the metric system often use the words mass and weight interchangeably. While scientifically the kilogram is only a unit of mass, in everyday life it is often used as a unit of weight as well.

Reasonable Values for Weight

To have an essential understanding of metric weight, you must be able to identify reasonable values for weight. When testing for reasonableness, you should assess both the unit and the unit value. Only by examining both can you determine whether the given weight is reasonable for the situation.

VIDEO

[Metric System: Units of Weight \(https://openstax.org/r/Metric_System:_units_of_weight\)](https://openstax.org/r/Metric_System:_units_of_weight)

EXAMPLE 9.31

Identifying Reasonable Units for Weight

Which is the more reasonable value for the weight of a newborn baby:

- 3.5 kg or
- 3.5 g?

Solution

Using our reference weights, a baby weighs more than 3.5 paperclips, so 3.5 kilograms is a more reasonable value for the weight of a newborn baby.

> YOUR TURN 9.31

1. Which represents a reasonable value for the weight of a penny:
2.5 g or 2.5 kg

EXAMPLE 9.32

Determining Reasonable Values for Weight

Which of the following represents a reasonable value for the weight of three lemons?

- 250 g,
- 2,500 g, or
- 250 kg?

Solution

Because a kilogram is about 2.2 pounds, we can eliminate 250 kg as it is way too heavy. 2,500 grams is equivalent to 2.5 kilograms, or about five pounds, which is again, too heavy. So, a reasonable value for the weight of three lemons would be 250 grams.

> YOUR TURN 9.32

1. Which represents a reasonable value for the weight of a car:
1,300 g, 130 kg, or 1,300 kg?

?? WHO KNEW?**How Do You Measure the Weight of a Whale?**

It is impossible to weigh a living whale. Fredrik Christiansen from the Aarhus Institute of Advanced Studies in Denmark developed an innovative way to measure the weight of whales. Using images taken from a drone and computer modeling, the weight of a whale can be estimated with great accuracy.

▶ VIDEO

[Using Drones to Weigh Whales? \(https://openstax.org/r/Using_drones_to_weigh_whales?\)](https://openstax.org/r/Using_drones_to_weigh_whales?)

EXAMPLE 9.33**Explaining Reasonable Values for Weight**

The blue whale is the largest living mammal on Earth. Which of the following is a reasonable value for the weight of a blue whale: 149 g, 149 kg, or 149 mt? Explain your answer.

✓ Solution

A reasonable value for the weight of a blue whale is 149 metric tons. Both 149 g and 149 kg are much too small a value for the largest living mammal on Earth.

> YOUR TURN 9.33

1. The Etruscan shrew is one of the world's smallest mammals. It has a huge appetite, eating almost twice its weight in food each day. Its heart beats at a rate of 25 beats per second! Which of the following is a reasonable value for the weight of an Etruscan shrew: 2 g, 2 kg, or 2 mt? Explain your answer.

Converting Like Units of Measures for Weight

Just like converting units of measure for distance, you can convert units of measure for weight. The most frequently used conversion factors for metric weight are illustrated in [Figure 9.13](#).

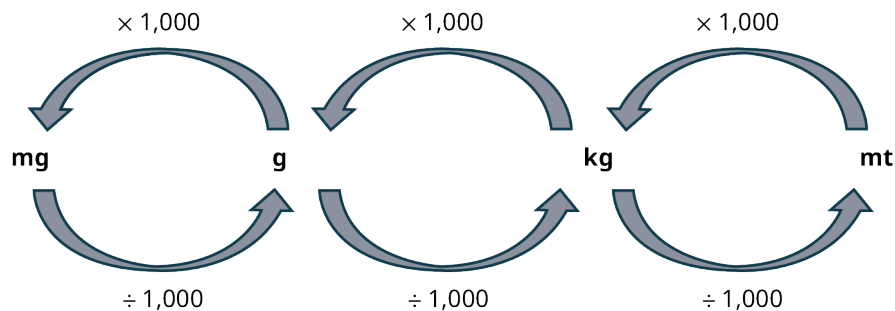


Figure 9.13 Common Conversion Factors for Metric Weight Units

EXAMPLE 9.34**Converting Metric Units of Weight Using Multistep Division**

How many kilograms are in 24,300,000 milligrams?

✓ Solution

Use division to convert from a smaller metric weight unit to a larger metric weight unit. To convert from milligrams to kilograms,

Step 1: Divide the value of the weight in milligrams by 1,000 to first convert from milligrams to grams.

Step 2: Divide by 1,000 again to convert from grams to kilograms.

$$\frac{24,300,000 \text{ mg}}{1,000} = 24,300 \text{ g}$$

$$\frac{24,300 \text{ g}}{1,000} = 24.3 \text{ kg}$$

So, 24,300,000 milligrams are equivalent to 24.3 kilograms.

YOUR TURN 9.34

1. How many kilograms are in 175,000 milligrams?

EXAMPLE 9.35

Converting Metric Units of Weight Using Multiplication

The average ostrich weighs approximately 127 kilograms. How many grams does an ostrich weigh?

Solution

Use multiplication to convert from a larger metric weight unit to a smaller metric weight unit. To convert from kilograms to grams, multiply the value of the weight by 1,000.

$$127 \text{ kg} \times 1,000 = 127,000 \text{ g}$$

The average ostrich weighs 127,000 grams.

YOUR TURN 9.35

1. The world's heaviest tomato weighed 4.869 kg when measured on July 15, 2020. How much did the tomato weigh in grams?

EXAMPLE 9.36

Converting Metric Units of Weight Using Multistep Multiplication

How many milligrams are there in 0.025 kilograms?

Solution

Use multiplication to convert from a larger metric weight unit to a smaller metric weight unit. To convert from kilograms to grams,

Step 1: Multiply the value of the weight by 1,000.

Step 2: Multiply the result by 1,000 to convert from grams to milligrams.

$$0.025 \text{ kg} \times 1,000 = 25 \text{ g}$$

$$25 \text{ g} \times 1,000 = 25,000 \text{ mg}$$

So, 0.025 kilograms is equivalent to 25,000 milligrams.

> YOUR TURN 9.36

1. How many milligrams are there in 1.23 kilograms?

> VIDEO

Metric Units of Mass: Convert mg, g, and kg (https://openstax.org/r/Convert_mg-g-kg)

Solving Application Problems Involving Weight

From children's safety to properly cooking a pie, knowing how to solve problems involving weight is vital to everyday life. Let's review some ways that knowing how to work with metric weight can facilitate important decisions and delicious eating.

EXAMPLE 9.37

Comparing Weights to Solve Problems

The maximum weight for a child to safely use a car seat is 29 kilograms. If a child weighs 23,700 grams, can the child safely use the car seat?

 Solution

Step 1: Convert the child's weight in grams to kilograms.

$$23,700 \text{ g} \div 1,000 = 23.7 \text{ kg}$$

Step 2: Compare the two weights.

$$23.7 \text{ kg} < 29 \text{ kg}$$

Yes, the child can safely use the car seat.

> YOUR TURN 9.37

1. The dosage recommendations for a popular brand of acetaminophen are listed in table below. What is the recommended dosage for a child who weighs 17,683 grams?

Weight	Dosage
11 kg to 15 kg	5 mL
16 kg to 21 kg	7.5 mL
22 kg to 27 kg	10 mL

EXAMPLE 9.38

Solving Multistep Weight Problems

A recipe for scones calls for 350 grams of flour. How many kilograms of flour are required to make 4 batches of scones?

 Solution

Step 1: Multiply the grams of flour need by 4 to determine the total amount of flour needed.

$$350 \text{ g} \times 4 = 1,400 \text{ g}$$

Step 2: Convert from grams to kilograms.

$$\frac{1,400 \text{ g}}{1,000} = 1.4 \text{ kg}$$

So, 1.4 kilograms of flour are needed to make four batches of scones.

YOUR TURN 9.38

1. A croissant recipe calls for 500 g of flour. How many kilograms of flour are required to make 10 batches of croissants?

EXAMPLE 9.39

Solving Complex Weight Problems

The average tomato weighs 140 grams. A farmer needs to order boxes to pack and ship their tomatoes to local grocery stores. They estimate that this year's harvest will yield 125,000 tomatoes. A box can hold 12 kilograms of tomatoes. How many boxes does the farmer need?

Solution

Step 1: Determine the total estimated weight of the harvested tomatoes.

$$140 \text{ g} \times 125,000 = 17,500,000 \text{ g}$$

Step 2: Convert the total weight from grams to kilograms.

$$17,500,000 \text{ g} \div 1,000 = 17,500 \text{ kg}$$

Step 3: Divide the weight of the tomatoes by the weight each box can hold.

$$17,500 \text{ kg} \div 12 \text{ kg} \approx 1,458$$

So, the farmer will need to order 1,458 boxes.

YOUR TURN 9.39

1. The average potato weighs 225 grams. A grocery chain orders 5,000 bags of potatoes. Each bag weighs 5 kg. Approximately how many potatoes did they order?

Check Your Understanding

For the following exercises, determine the most reasonable value for each weight.

25. Candy bar:
50 kg, 50 g, or 50 mg
26. Lion:
180 kg, 180 g, or 180 mg
27. Basketball:
624 kg, 624 g, or 624 mg

For the following exercises, convert the given weight to the units shown.

28. 8,900 g = _____ kg

29. $17 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$

30. $0.07 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$

For the following exercises, determine the total weight in the units shown.

31. three 48 g granola bars $\underline{\hspace{2cm}} \text{ kg}$

32. seven 28 g cheese slices $\underline{\hspace{2cm}} \text{ mg}$

33. six 15 mg tea bags $\underline{\hspace{2cm}} \text{ g}$



SECTION 9.4 EXERCISES

For the following exercises, determine the most reasonable value for each weight.

1. Aspirin tablet:
300 kg, 300 g, or 300 mg
2. Elephant:
5,000 kg, 5,000 g, or 5,000 mg
3. Baseball:
145 kg, 145 g, or 145 mg
4. Orange:
115 kg, 115 g, or 115 mg
5. Pencil:
6 kg, 6 g, or 6 mg
6. Automobile:
1,300 kg, 1,300 g, or 1,300 mg

For the following exercises, convert the given weight to the units shown.

7. $3,500 \text{ g} = \underline{\hspace{2cm}} \text{ kg}$
8. $53 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$
9. $0.02 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
10. $200 \text{ mg} = \underline{\hspace{2cm}} \text{ g}$
11. $2.3 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$
12. $20 \text{ kg} = \underline{\hspace{2cm}} \text{ mg}$
13. $2,300 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
14. $8,700 \text{ mg} = \underline{\hspace{2cm}} \text{ g}$
15. $9,730 \text{ mg} = \underline{\hspace{2cm}} \text{ kg}$
16. $0.0078 \text{ kg} = \underline{\hspace{2cm}} \text{ g}$
17. $2.34 \text{ g} = \underline{\hspace{2cm}} \text{ mg}$
18. $234.5 \text{ mg} = \underline{\hspace{2cm}} \text{ g}$

For the following exercises, determine the total weight in the units shown.

19. Three 350 mg tablets $\underline{\hspace{2cm}} \text{ g}$
20. Seven 115 g soap bars $\underline{\hspace{2cm}} \text{ kg}$
21. Six 24 g batteries $\underline{\hspace{2cm}} \text{ mg}$
22. Fifty 3.56 g pennies $\underline{\hspace{2cm}} \text{ mg}$
23. Eight 2.25 kg bags of potatoes $\underline{\hspace{2cm}} \text{ g}$
24. Four 23 kg sacks of flour $\underline{\hspace{2cm}} \text{ g}$
25. Ten 2.5 kg laptops $\underline{\hspace{2cm}} \text{ g}$
26. Seven 1,150 g chickens $\underline{\hspace{2cm}} \text{ kg}$
27. Ninety 4,500 mg marbles $\underline{\hspace{2cm}} \text{ kg}$
28. There are 26 bags of flour. Each bag weighs 5 kg. What is the total weight of the flour?
29. The average female hippopotamus weighs 1,496 kg. The average male hippopotamus weighs 1,814 kg. How much heavier, in grams, is the male hippopotamus than the female hippopotamus?
30. Twelve pieces of cardboard weigh 72 grams. What is the weight of one piece of cardboard?
31. Miguel's backpack weighs 2.4 kg and Shanayl's backpack weighs 2,535 grams. Whose backpack is heavier and by how much?
32. A souvenir chocolate bar weighs 1.815 kg. If you share the candy bar equally with two friends, how many grams of chocolate does each person get?